

TensorRT optimises the trained neural network to get an optimised inference engine, and makes it possible to train AI machines efficiently.

AI medical assistance apps

Many apps linked with AI systems can be helpful during a pandemic. AI based apps that can give medical assistance and keep track of infected individuals are useful in such situations. There are many cloud based apps available that can give medical assistance but without AI, such apps won't be able to predict the worst-case scenarios during a pandemic. AI based apps can improve themselves and find better ways to handle newer situations. Such medical assistance apps can give appropriate guidance to the patients according to the level of severity, and some apps with chatbots will even be able to communicate with every individual without the need of any human intervention. The Software as a Service (SaaS) cloud system with AI can be used to give such medical assistance with the help of smartphone apps or any Web based application.

During the COVID-19 pandemic in China, apps like Jingdong and Wei Yi provided online medical consultation services to patients. These apps work through a cloud based AI system. Some AI apps like KardiaMobile 6L and TeleICU were also used during the pandemic to monitor patients.

A company called AliveCor has launched KardiaMobile 6L, an AI-enabled platform, that allows healthcare professionals to measure QTc (heart rate corrected interval) in COVID-19 patients. The QTc measurement can help to determine the probability of sudden cardiac arrest. The AI platform has been developed based on the FDA's recent guidance to allow more availability of non-invasive remote monitoring devices for the pandemic. Another company, called CLEW, has launched the TeleICU

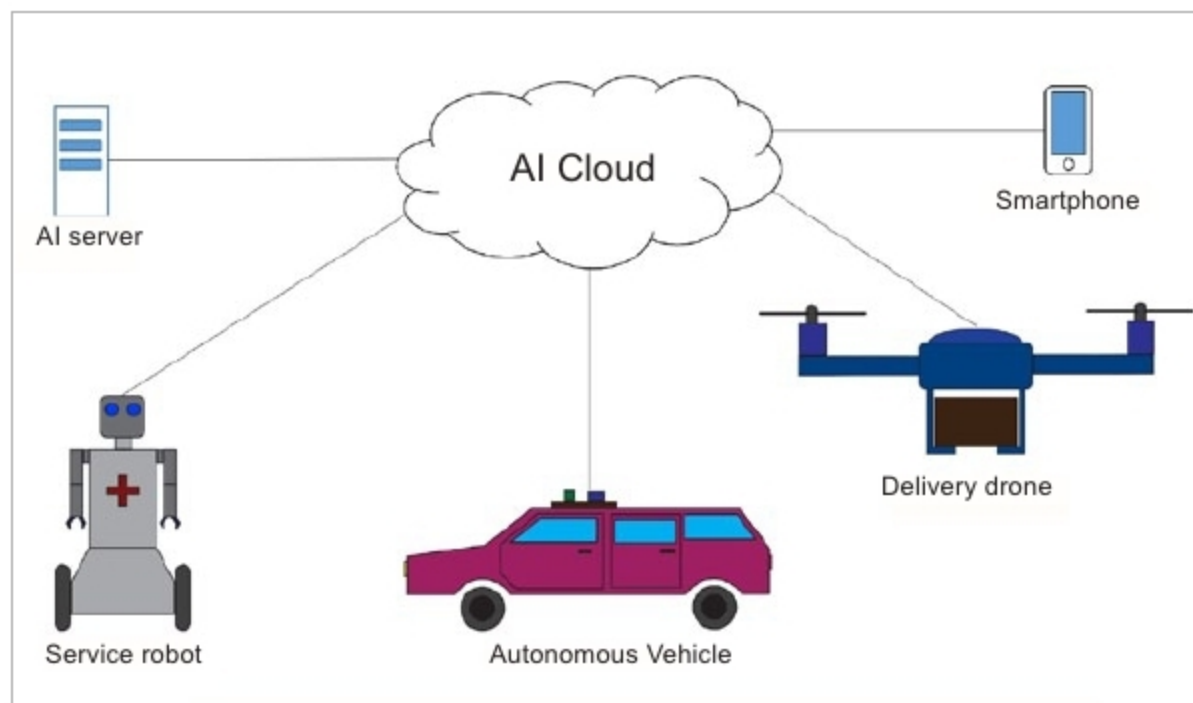


Figure 6: Collaborative AI for autonomous machines

and uses AI to identify respiratory deterioration in advance. Pneumonia is the most common complication of the COVID-19 infection, but with the help of AI, it's now possible to detect pneumonia by analysing a CT scan in less than 60 seconds, with accuracy as high as 92 per cent and a recall rate of 97 per cent on test data sets. This was made possible in China by an open source AI model that analysed CT images and not only identified lesions but also quantified them in terms of number, volume and proportion.

Therefore, with the help of FOSS tools, it's possible to create such AI medical apps that can help during a pandemic.

The use of autonomous robots, vehicles and drones during a pandemic

Using service robots can also help people during any pandemic. As robots do not get infected, they can be deployed to help people with any kind of infection. During the COVID-19 pandemic, hospitals in China at Guangdong province used service robots to deliver medicines and communicate with the infected patients. Autonomous vehicles are also useful during any pandemic because they do not need a human driver. Such vehicles use varieties of

sensors like radar, lidar, sonar, GPS, odometry and inertial measurement units, along with AI algorithms, to drive from one place to another autonomously. During any pandemic, public transport gets suspended to prevent the spread of infection. In such a crisis, autonomous vehicles can be useful for the transfer and delivery of essential goods.

Autonomous drones can also be used to deliver essential goods like food and medicines to people during a pandemic. Such drones use AI and GPS to auto-navigate, and can deliver goods to a predefined location. Drones can also be used for the surveillance of huge gatherings and monitor restricted places during a pandemic.

All such autonomous machines like service robots, vehicles and drones can work together using a collaborative development model. This will enable autonomous machines to learn and exchange data efficiently using an AI cloud. With the help of many FOSS tools like TensorRT and hardware like Nvidia AGX, it's possible to create efficient autonomous machines that can help during a pandemic. **END** 🐧

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